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Review

Utilizing Different Crop Rotation Systems for Agricultural and Environmental Sustainability: A Review

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Abstract

Monoculture involves growing the same crop on the same land over at least two crop cycles. Continuous monoculture can increase the population density of pests and pathogens over time, thereby reducing agricultural yields and increasing dependence on chemical inputs. Crop rotation is an agricultural practice that involves systematically and sequentially planting different crops in the same field over multiple growing seasons. This review explores the advantages of crop rotation and its contribution to promoting sustainable farming practices, such as legume integration and cover cropping. It is based on a thematic literature review of peer-reviewed studies published between 1984 and 2025. We found that crop rotation can significantly improve soil structure and organic matter content and enhance nutrient cycling. Furthermore, soil organic carbon increased by up to 18% when legumes were included in rotations compared to monoculture systems in Europe, while also mitigating greenhouse gas emissions, enhancing carbon sequestration, and decreasing nutrient leaching and pesticide runoff. Farmers can adopt several strategies to optimise crop rotation benefits, such as diversification of various crops, legume integration, cultivation of cover crops, and rotational grazing. These practices ensure agricultural sustainability and food security and support climate resilience.

Keywords: crop rotation; greenhouse gas emissions; monoculture; soil fertility; sustainable practices

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Received: 9 July 2025
Revised: 7 August 2025
Accepted: 10 August 2025
Published: 14 August 2025

Citation: Al-Musawi, Z.K.; Vona, V.; Kulmány, I.M. Utilizing Different Crop Rotation Systems for Agricultural and Environmental Sustainability: A Review. *Agronomy* **2025**, *15*, 1966. https://doi.org/10.3390/agronomy15081966

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1. Introduction

The global demand for sustainable agriculture has increased in response to the growing awareness of the environmental impact of unsustainable agricultural practices, particularly climate change, biodiversity loss, and ecological degradation [1–3]. For example, the global organic market reached USD 135 billion in 2022, and by 2022, organically managed land had increased by more than 65% compared to 2012 [4–6]. Additionally, the global adoption of conservation agriculture, including practices such as crop rotation, no-till, and cover cropping, has exceeded 205 million hectares as of 2023, accounting for nearly 15% of global arable land [7]. Sustainable farming methods aim to optimise agricultural productivity while preserving the environment. Therefore, it is essential to explore practices that reduce greenhouse gas (GHG) emissions and maintain soil health [8].

Crop rotation is a central component of sustainable agriculture, referring to a planned sequence of different crops grown in the same field across seasons to improve soil health and productivity. It differs from other agricultural practices, such as intercropping and monoculture (monocropping) [9]. Crop rotation has been a fundamental practice in many civilisations. Historical records from ancient Rome indicate that farmers recognised the benefits of alternating crops for restoring soil quality and health and breaking pest cycles and performed an early form of rotational cropping based on food, feed, and fallow sequences [10]. The 'three-field method', a type of crop rotation, was used in the Middle Ages to boost agricultural output in Europe, and research on the effects of crop rotational diversity on yields and soil health is still ongoing [11–13]. Over the centuries, rotation systems became increasingly structured, particularly as farming practices intensified during the agricultural revolution of the 18th and 19th centuries. Crop rotation has developed into a systematic approach to improve yields and reduce soil depletion, with farmers rotating crops such as legumes that form symbiotic relationships with nitrogen-fixing microorganisms to naturally restore soil nutrients [14,15]. In the early 20th century, in the Southern United States, George Washington Carver advocated for improving soil fertility by rotating cotton with peanuts and other legumes, demonstrating how crop rotation could restore degraded soils and reduce dependency on synthetic inputs [16]. In modern agriculture, crop rotation remains an effective strategy for balancing productivity and environmental stewardship, particularly in regions heavily affected by industrial farming [17]. It plays a critical role in improving climate resilience, minimising dependence on synthetic inputs, enhancing soil carbon sequestration, and supporting global goals for sustainable intensification and food security [7,18].

Currently, crop rotation systems vary according to weather conditions. For example, in the U.S. Midwest, corn and soybean rotation is prevalent, where soybeans fix nitrogen in the soil, which helps subsequent corn crops [19]. Farmers in Southeast Asia commonly rotate rice with legumes, such as rice–mung bean or rice–vegetable legumes, thereby enhancing nitrogen content and decreasing pest pressure [20].

Compared with other practices, such as monoculture, crop rotation presents many benefits, such as improving soil structure, quality, and health, as well as reducing carbon dioxide (CO_2) emissions [18], thus mitigating some of the adverse environmental effects of agriculture [16]. This practice has evolved as a viable method to reduce dependence on chemical fertilisers to enhance soil nutrition and promote crop yield, reduce GHG emissions [15], and control crop pests [21]. This review represents a thematic synthesis of crop rotation research between 1984 and 2025, covering different agroecological systems and integrating agronomic, environmental, and implementation perspectives. It also addresses research gaps, including limited site-specific information under climate variability, limited long-term trials in tropical systems, and a lack of socioeconomic analysis of adoption in smallholder farming. This review first identifies the most promising crop rotation approaches, then compares their advantages and disadvantages with those of other cropping systems, and finally examines their practical application, thus highlighting the value of crop rotation and its practical importance in sustainable agriculture.

2. Materials and Methods

The data for this literature review were obtained by conducting extensive searches across various bibliometric databases, including Scopus, Cambridge Journals, MDPI, PubMed, Web of Science, Taylor & Francis, ScienceDirect, Springer, and AGRO. These databases were specifically chosen for their extensive archives of peer-reviewed academic journals in the agricultural and environmental sciences. This approach ensured that the

sourced articles were of high academic and scientific value and directly relevant to our research themes.

The analysis excluded articles published in languages other than English to ensure consistency in data interpretation, as well as popular science articles and those describing cellular-level research findings. This review focused on research papers published between 1984 and 2024. The articles selected for the review are discussed in the thematically relevant sections of the manuscript. Search terms included combinations of keywords, including 'Crop rotation', 'Greenhouse gas emissions', 'Monoculture', 'Soil fertility', and 'Sustainable practices'. This specific selection of keywords aimed to encompass a broad spectrum of research topics within the scope of agricultural practices and their environmental impacts.

Over 300 studies were initially identified and screened for their relevance. After screening for relevance, duplication, and language, 226 studies met the inclusion criteria and were thoroughly analysed and included in the final review. To categorise the results systematically, studies were grouped according to the specific agricultural practice(s) investigated (crop rotation in agriculture and the environment). The classification was based on key variables reported in the literature, including cropping systems, agronomic outcomes (e.g., yield and nutrient cycling), environmental metrics (e.g., GHG emissions, soil health), and implementation strategies. When possible, papers were further stratified by crop type (e.g., cereals and legumes), specific crop studied (e.g., maize, wheat, and soybeans), geographic region, soil type, and climate, providing a comprehensive overview that reflects diverse conditions and practices across the globe.

3. Crop Rotation Strategies

3.1. Crop Rotation

Crop rotation is a key practice in sustainable farming [22] and can reduce pests, improve soil health, and increase crop yield if implemented strategically. An important aspect of crop rotation is the selection of appropriate crops to alternate each season to disrupt pest development cycles and enhance nitrogen accumulation [23]. For example, crop rotation alters the life cycle of insects and pathogens by hindering their reproduction. Crop rotations that include clover and rotations of wheat and legumes are beneficial for improving soil fertility. Legumes help to build nitrogen in the soil. Therefore, crop rotation is not typically chosen in industrial agriculture to conserve nitrogen, as these systems depend heavily on synthetic nitrogen fertilisers to achieve high yields, making natural nutrient cycling less essential [24–26].

Additionally, crop rotation with crops with different root structures and nutrient requirements can significantly alter soil physicochemical properties. For example, corn–wheat–soybean rotation has been cited for its agronomic benefits. Wheat utilises nitrogen efficiently, soybeans improve its availability through nitrogen-fixing bacteria, and corn enhances biomass and utilises the remaining nitrogen. These effects are, in principle, supported; however, more region-specific field studies are needed to quantify their long-term impact under different agroecological conditions. This rotation improves soil quality by increasing the soil mass, organic carbon, microbial biomass, and nutrient levels, while lowering the pH [27,28]. This improves production while reducing the need for synthetic fertilisers and pesticides.

Crop rotation is affected by the climate and type of soil in each area. Crop rotation is primarily practiced in colder climates, such as in Finland, using wheat or barley to limit pests and improve soil conditions [19,29]. Corn–soybean rotations are frequently implemented in the American Midwest because the nitrogen requirements for corn can be met by deposits from prior soybean planting [19].

Tropical and subtropical areas, such as Southeast Asia, have rice–legume vegetable systems, which help eliminate pests related to rice and legumes and improve soil fertility through legumes. Area sensitivity is critical because the climatic conditions of a given area and soil type determine the effectiveness of crop rotation. For example, desert areas should switch between plant species that can tolerate the effects of climate change on rainfall and temperature patterns, such as drought and heat waves. Contrastingly, zones characterised by abundant rainfall should be used for rice cultivation [20,30].

3.2. Legume Integration

The inclusion of legumes in crop rotations increases nitrogen cycling, significantly influencing plant growth [31]. Bean, clover, and alfalfa contain root nodules with symbiotic bacteria that can fix nitrogen from the atmosphere into bioavailable forms [31,32]. These bacteria convert atmospheric nitrogen (N_2) into ammonia (NH_3) through a complex enzymatic process involving nitrogenase. The ammonia is then converted into other forms, such as ammonium (NH_4^+), which can be easily absorbed by plants and used for growth [33,34]. This facilitates natural nitrogen fixation, reducing the need for artificial nitrogen fertilisers and the associated costs and environmental impacts [35]. The amount of nitrogen that is fixed by legumes varies depending on their species, environmental conditions, and agriculture management practices. In the case of soybean in the Midwest, the upper limit for average nitrogen fixation is estimated at approximately 75 kg N/ha. This estimation assumes that soybean plants in the Midwest accumulate a total of 150 kg N/ha, with approximately 50% of this amount being supplied by biological nitrogen fixation [36,37]. Alfalfa, which widely grown in temperate regions, has been shown to fix approximately 148 kg N/ha during the growing season [38]. Additionally, adding clover or alfalfa into rotations can benefit crops by improving the structure, nutrient content, and water-holding capacity of the soil. Organically managed legume crops such as lentil or vetch are often inserted into a rotational system to maintain soil fertility without recourse to chemical fertilisers [39,40]. Besides the fixed nitrogen, legumes can also improve the availability of phosphorus in soil, improve soil structure, and increase biodiversity. Studies have found that legumes can increase the amount of organic matter and enhance water infiltration rates, leading to improved soil health and reduced erosion. Legume crop rotations promote soil health and reduce reliance on synthetic inputs, contributing to long-term agricultural sustainability [41].

3.3. Cover Cropping

Cover crops provide soil protection, including improving soil texture and nutrient conservation [42]. They conserve the soil by anchoring the soil particles with their root systems, thereby preventing erosion by wind or water [43]. The sustainable use of cover crops also aids in soil and water conservation by adding to the soil organic matter (SOM) content, which leads to improved water and root-holding capacities required for subsequent crop production [44]. While accurate global data may vary, the adoption of cover crops by farmers is increasing. According to a survey conducted between 2014 and 2015 in the United States, cover crop planting has shown a steady increase, averaging 105 hectares (259 ac) per farm. This is further substantiated by the findings of the 2012 Agriculture Census report, as 4.2 million ha (10.3 million ac) of cover crops were planted. This demonstrates that if current trends continue, this figure could exceed 8 million ha (20 million ac) after approximately 8 years [45].

Common cover crops include legumes, such as clover and vetch; grasses, such as rye and barley; and *Brassica* and Raphanus species, such as mustard and radish, with each presenting distinct agronomic benefits. For instance, rye is effective in suppressing weeds

through multiple mechanisms including competitive effects and allelopathic compounds, while mustard is used as a biopesticide to control soil-borne pests and demonstrates high competitiveness as a cover crop species [46,47]. Rye cover crops limit weeds through resource competition, releasing natural weed-inhibiting chemicals, creating dense canopies that reduce light, and producing residues that further continue suppression [48]. Likewise, mustard cover crops grow quickly, compete with weeds for resources, and release allelopathic compounds like glucosinolates [49,50]. For example, it has been observed that nematode and soilborne pathogens populations are substantially reduced in fields containing cover crops, such as mustard, particularly in those in the *Brassica* genus, in temperate regions such as Lower Saxony, Germany. This effect is likely due to the release of biofumigant compounds, such as glucosinolates. Similarly, in Germany and the United States, it has been observed that rye is capable of supressing weeds under different farming systems [47]. Among the most significant advantages of cover cropping is the enhancement of soil fertility [51]. Leguminous cover crops such as clover and vetch biologically fix nitrogen, rendering atmospheric nitrogen available for subsequent crops. Results from studies have demonstrated that the amount of nitrogen fixed by legume cover crops in the soil can range from 10 to 217 kg N/ha over a single growing season, depending on the species used and the site-specific conditions [52]. In field trials conducted in North Carolina, United States, hairy vetch fixed up to 217 kg N/ha over a single season (e.g., cultivar 'PRO' at Piedmont, late termination), while lupin (TifBlue78) and subterranean clover (Denmark) contributed as little as 15 and 16 kg N/ha, respectively, within the same timeframe. Similarly, it was determined in a long-term no-till cropping experiment in Kansas, United States, that Sunn hemp, a high-biomass summer legume, fixed up to 172 kg N/ha within 14 weeks, increasing total soil nitrogen by up to 279 kg N/ha compared with non-cover cropped plots [53]. The nitrogen that is fixed by these crops can substantially offset synthetic fertiliser needs, especially in low-input systems. It is an economical and natural process that reduces synthetic fertiliser usage [54]. The organic material from decomposing cover crops further enhances soil structure and increases water retention, aeration, and microbial activity [47,55].

Cover cropping is also a fundamental erosion control method. Cover crop root systems stabilise the soil and reduce wind and water erosion on sloping or high-precipitation land. Studies have shown that using cover crops can decrease soil erosion by 11–29% under Midwestern cropping systems [56]. Moreover, field experiments conducted in the southeastern United States demonstrated a 64% reduction in erosion rates when compared to bare soil [57]. Additionally, Olson et al. [58] mentioned a soil erosion level of 17.9 tons/acre/year in non-controlled systems; Langdale et al. [59] noted that cover crops reduced soil loss from 30.1 to 24.3 $Mg \cdot ha^{-1} \cdot year^{-1}$. Kaspar and Singer [60] observed significant erosion reductions in a variety of cropping systems with cover crop adoption. By incorporating cover crops into crop rotation systems, producers can enhance soil health, stabilise yields, and positively contribute to long-term agricultural sustainability [61,62]. Moreover, using cover crops can lead to similar or even higher crop yields, while enhancing their durability. In the United States, cover crops have been shown to reduce crop insurance losses from drought, high temperature, and high level of moisture, reflecting enhanced resilience under climatic stress [63]. Additionally, winter legume cover crops have found to significantly increase corn yields by an average of 24% compared to crops that do not use cover crops. Moreover, a 37% yield increase was observed when no nitrogen fertiliser was used, whereas grass cover crops-maintained yields at levels comparable to those of no-cover systems [64].

3.4. Incorporation of Livestock

Rotational grazing involves the integration of livestock rearing and crop rotation to promote soil health and pasture productivity. Livestock are moved to various sections of a field where they feed, allowing the crops in previously grazed areas to recover [65]. Galindo et al. [65] noted that over a four-year period, rotational grazing in Minnesota and Pennsylvania greatly improved soil organic carbon (SOC) and enhanced enzymatic activity, specifically, β-glucosidase, alkaline phosphatase, and arylsulfatase, indicating improved nutrient cycling and enhanced enzymatic activity.

Animal manure introduces organic matter and nutrients into the soil, which benefits soil bacteria and overall soil health [66,67]. In fact, consistent rotational systems in grazing have been shown to enhance nitrogen use efficiency and store more carbon in the soil by approximately 0.3–1.6 Mg C ha^{-1} yr^{-1}, depending on local conditions and management practices [67]. This aligns with field results for regenerative systems in Spain, where rotational grazing enhanced topsoil carbon compared to conventional methods, especially with longer rest periods [68]. It is also supported by global meta-analyses highlighting that rotational grazing increases SOC compared to continuous grazing [69].

This management practice optimises the utilisation of forage crops, avoids overgrazing, and encourages pastureland restoration by allowing rest and recovery between grazing periods. In the Midwest and semi-arid Texas, U.S, this system has been shown to enhance soil biological activity and physical resilience [65,70]. By regulating grazing intensity and timing, rotational grazing minimises the adverse effects of continuous grazing, such as soil compaction, plant biodiversity loss, and nutrient depletion [71]. Furthermore, rotating livestock contributes to effective weed and pest control, thus minimising the need for chemical controls. It also prevents soil erosion and maintains vegetation variety, supporting valuable insect species for pollination and pest control, thereby maintaining the biological diversity essential for a resilient ecosystem [72]. Rest periods between short grazing intervals allow plant species to regrow and root systems to stabilise, improving soil aeration and water retention. According to studies conducted in Iowa, U.S, there were no significant differences observed in SOM when exposed to short-term grazing periods. However, long-term grazing in Virginia and the southeastern United States reported increases of 22–31.9% in SOC with management-intensive grazing [71,73–75].

Rotational grazing enhances the growth of deeper and more vigorous root systems, which improves soil structure, nutrient cycling, and erosion resistance. Improved root mass increases the carbon sequestration potential of pastures, reducing atmospheric CO_2 concentrations [65]. Adaptive multi-paddock grazing systems enhance water infiltration and boost soil microbial biomass, fungal-to-bacterial ratios, and overall microbial function, contributing to healthier soil conditions and ecological resilience [65]. A previous study demonstrated a 20–60% rise in infiltration rates and improvements in microbial biomass under adaptive grazing management [76,77]. Integrating livestock into crop cycles, particularly on heterogeneous farms with diverse crops and landscapes, promotes mutualism between plants and animals, enhancing soil fertility and reducing vulnerability to pest damage and temperature extremes. Additionally, cover crops can cut down erosion by up to 90% and reduce pesticide runoff by more than 50%, while providing up to 132 kg N/ha for subsequent crops [78].

Controlled manure deposition through rotational grazing increases microbial activity and nutrient cycling. This reduces the need for synthetic fertilisers and supports sustainable fertility management through improved nutrient uptake efficiency and reduced environmental loss [79]. Integrated manure management also prevents leaching and nutrient runoff, mitigating water pollution risks [67,80].

Animals grazing on crop residues during harvest or on cover crops in off-cropping seasons can sustainably recycle nutrients for crops and livestock [72,80]. Using combined grazing methods such as those incorporating forage legumes and managed grazing, such as those employed in Florida and southern Brazil, has been shown to improve nutrient cycling and decrease the need for synthetic fertilisers, without compromising productivity [80]. Furthermore, rotational grazing reduces parasitic infestation and disease transmission among livestock [81] and improves livestock gut health, potentially lowering veterinary costs [82]. According to van der Voort et al. [81], using better grazing methods help reduce parasites, boosts productivity, and saves USD 9.03 per cow each year.

Increased plant species diversity improves ecological stability and ensures long-term viability of livestock systems. By reducing dependency on external inputs and encouraging the natural regenerative capacity of pasture systems, rotational grazing exemplifies sustainable agriculture. It is consistent with ecological guidelines and focuses on conservation, yielding results that benefit nature and the economy [70,83].

3.5. Alternate Intercropping Systems

Changing intercropping or strip intercropping with annual alternation of crops in adjacent strips combines advantages of intercropping and crop rotation. It promotes complementarity across spatial and temporal niches, minimises limitations of continuous cropping challenges, and enhances soil health as well as supporting dual harvests within a single year. In cotton–peanut rotations, this approach has been reported to increase yields by 17–21%, net returns by 10–23%, and land equivalent ratios (LERs) by 20–30% [84]. In cotton–soybean systems, the modification of planting dates and selection of appropriate varieties mitigate interspecific competition, achieving LERs between 1.04 and 1.15 and promoting canopy photosynthesis and root growth [85,86]. These systems offer high yield resilience and ecological efficiency, placing them as integral components of sustainable intensification strategies. A summary of the crop rotation methods discussed in Sections 3.1–3.5 is presented in Table 1.

Table 1. Summary of crop rotation methods and their impacts (Sections 3.1–3.5).

Rotation Strategy	Main Advantages	Quantitative Representative Outcomes
3.1 General Crop Rotation	Disrupts pest/disease cycles; improves soil health and yield; enhances nitrogen cycling; improves soil organic carbon (SOC) and pH balance.	Up to 18% increase in SOC; yield increase of 13–44% in cereal–legume sequences; 39% reduction in N_2O emissions.
3.2 Legume Integration	Fixes atmospheric nitrogen (e.g., soybean ~75 kg N/ha, alfalfa ~148 kg N/ha); enhances P availability; boosts soil biodiversity and fertility.	Soybean biological nitrogen fixation (BNF) ~75 kg N/ha; alfalfa ~148 kg N/ha; improved nutrient cycling and organic matter; enhances microbial activity.
3.3 Cover Cropping	Reduces erosion by 11–29%; increases soil organic matter (SOM); fixes up to 217 kg N/ha (e.g., hairy vetch); suppresses weeds via competition and allelopathy.	Hairy vetch fixed 217 kg N/ha; erosion reduced by 11–29%; crop yields increased up to 24% (corn); reduced pesticide runoff by >50%.
3.4 Rotational Grazing	Increases SOC by 0.3–1.6 Mg C/ha/year; improves enzymatic activity; enhances water retention and microbial biomass; reduces input needs.	SOC gains 0.3–1.6 Mg C/ha/year; enzymatic activity (β-glucosidase, phosphatase) increased; 22–32% SOM rise in long-term trials.
3.5 Alternate Intercropping	Increases land equivalent ratios (LERs (20–30%)); improves root–shoot interactions; enhances canopy photosynthesis and yield stability.	Yield increase: cotton-peanut 17–21%; LER improvement up to 30%; increased economic return by 10–23%; improved stress resilience.

4. Advantages and Disadvantages of Crop Rotation

4.1. Soil Fertility

Crop rotation and monocropping systems are often compared because they significantly differ in their effects on soil fertility, nutrient cycling, and sustainability [87]. In the following section, both systems are discussed to identify the specific benefits and drawbacks of crop rotation.

Crop rotation is often contrasted with monocropping regarding its long-term impact on soil and environmental health. These systems differ in their impact, use, carbon storage, and biodiversity outcomes (Table 2). Crop rotation plays a significant role in SOM accumulation. The decomposition of different plant residues stimulates the development of humus, a stable form of organic matter, resulting in enhanced microbial activity and soil structure through the provision of a food source and stabilising agent for soil particles [88–90].

According to recent studies, root–shoot dynamics is one of the key mechanisms enhancing soil fertility in diversified systems. Improved root structure in rotations and intercropping increases nutrient uptake, root exudation, and microbial activity, which enhances nutrient cycling and soil structure [84,91]. These systems also result in improved adaptation to poor soils through more efficient nutrient foraging [92].

A diverse soil microbiome ensures improved nutrient cycling in addition to the suppression of soil-borne pathogens by fostering competition between microorganisms and producing compounds that inhibit pathogen growth, resulting in improved crop health [93–96].

Conversely, continuous monocropping can result in the exhaustion of specific nutrients like nitrogen, phosphorus, and potassium and establishment of a favourable environment in the soil for pests and diseases. By adopting crop rotation, farmers can break pest and disease cycles, reduce their frequency, and minimise the use of chemical pesticides, thereby protecting soil health and reducing environmental pollution [97–99].

While crop rotation has many advantages, it also has some drawbacks, including increased management complexity, more planning, and reduced compatibility with large-scale, single-crop machinery. Furthermore, it has the potential to result in lower short-term profitability compared to monocropping systems specialised for high-value cash crops [100–102]. For instance, in the Indo-Gangetic Plain, many small farmers avoid rotating between rice and legumes because of low market returns, despite soil benefits [35,103]. Additionally, while in the U.S. Midwest, corn–soybean is still the most common rotation [104], concerns about equipment needs in diversified systems are also being addressed; these systems often face additional costs or machinery incompatibility, with research showing improved mechanisation compatibility and specialised machinery development [105]. In dry sub-Saharan Africa, legume rotation adoption is limited by socio-economic barriers rather than by water scarcity. However, despite the long-term agronomic benefits, crop options are still limited by low water availability and lack of affordable irrigation [106,107].

Table 2. Comparison of single cropping and crop rotation systems.

Aspect	Monocropping	Crop Rotation
Soil Health and Fertility	Lower soil health index [108,109].	Improves soil health by 45%; adds organic matter [108,109].
	Nutrient depletion reduced microbial diversity [102,110].	Improves soil fertility, biodiversity [102,110].
	Degradation implied [111].	Stabilises yield across climates [111].
Microbial and Soil Biodiversity	Lower bacterial diversity [112,113].	Higher diversity in rotations like maize–alfalfa [113].
	Reduces biodiversity [108,110].	Enhance biodiversity [108,110]

Table 2. *Cont.*

Aspect	Monocropping	Crop Rotation
Soil Organic Carbon	Reduced soil organic carbon [109].	Significantly increases SOC [109]. Increased SOC by 18% [114]
Nutrient Management	Greater reliance on synthetic inputs; up to 120 kg N/ha in cereals [115].	Reduces N input: oats = 82 vs. clover–oats = 42 kg N/ha [115].
	Nitrogen limitation in corn-after-corn [111].	Soybean N fixation boosts corn yield [111].
	−41 kg N, −6 kg P, −26 kg K ha^{-1} yr^{-1} lost; higher fertiliser use [87].	Enhances nutrient cycling, esp. with legumes [87].
		Reduces N requirement [116].
GHG Emissions	Higher N_2O emissions [109].	Reduces N_2O emissions in rotation systems; one study reported a 39% decrease [109].
Climate Resilience	More vulnerable to climate variability [117].	Up to 1000 kg/ha more during heat/drought [117].
		Improves resilience to biotic/abiotic stress [116].
Sustainability and Environment	Associated with land degradation [118].	Promotes sustainable land use [118].
	Environmental degradation and deforestation [108].	Promotes sustainability and biodiversity [108].
Pest and Disease Management	Increased pest/disease pressure [102,108].	Reduces pest/disease pressure [102,116].
	Higher pest pressure [111].	Benefit increases under high pest pressure [111].
	Weed pressure may increase [26].	Suppresses weeds with rye, radish [26].
	High dependency on synthetic inputs [108].	Improves natural fertility, pest control [108].
Management and Scalability	Simpler and widely adopted [87,102].	Requires complex planning [87,102].
	Treated as control [111].	Treated as treatment [111].
	Traditional practice, being replaced [110].	Core principle of CA, increasingly adopted [110].
	Dependent on synthetic inputs [26].	Reduces input reliance; improves stability [26].
Economic Performance	Lower gross margins [119].	A 48% higher margin in faba bean–wheat rotation [119].
	More profit fluctuations [115]	Stable profits under variable weather [115].
	Requires more external input [110].	Decreases variability in returns [116].
		Reduces costs by 15–16% [110]. Economic outcomes may vary for small-scale farmers depending on local conditions and resource access.
Agrobiodiversity		Enhances agroecosystem diversity [116].
Food Safety	Higher risk of nitrate accumulation in edible crops because of excessive use of synthetic fertiliser and decrease nutrient efficiency in monocultures [108,120].	Potential contamination risk when manure managed poorly but otherwise supports to lower nitrate accumulation because of enhancing nutrient cycling and organic management systems [108,121].
Mechanization and Machinery Cost	Mechanisation-friendly machinery. Costs USD 67,200 [100].	Less compatible with single-crop machinery. Costs USD 115,000 [100].
Yield	Lower yields in cereals [117].	Higher yields; 860 kg/ha (winter), 390 kg/ha [117].
	Short-term yields maintained, long-term decline [87].	Stabilises yield via fertility and pest control [87].
	Lower yields under monoculture [111].	Corn +1.03 t/ha; Soy +0.21 t/ha higher under rotation vs. monoculture [111].
		Increased output in trials [26].

Note: The "Soil Health Index" refers to a composite measure typically based on physical, chemical, and biological soil indicators, which vary by study and region.

Different residues and root structures are deposited in the soil each time a crop is grown. Thereby, crop rotation improves the breakdown of SOM, increases soil carbon content, improves nutrient availability, and promotes the health of soil microorganisms [122,123]. Enhanced microbial populations and SOM improve soil structure, water retention, and aeration [124,125]. Different crops remove and replenish nutrients at varying

rates, preventing the long-term deficiency of specific elements. Crop rotation is therefore a natural method of enriching the nitrogen content of soils and reducing the need for synthetic fertiliser [126].

Various crop rotation methods, especially those that include diverse plant families, help maintain a balanced soil environment and promote the activity of certain microbes and fungi that benefit soil health by suppressing pathogens and enhancing nutrient cycling [127,128]. Sometimes plant roots or soils are synthetically inoculated with beneficial microbes; this is often referred to as 'labelling crops with microbes', by doing this, farmers are improving soil health. Moreover, crop rotations using crops such as corn, wheat, and soybean aid in maintaining the soil nitrogen balance, alternating nitrogen-demanding and nitrogen-fixing plants. This reduces nutrient depletion, enhances nutrient use efficiency, and improves soil productivity, thereby providing better plant growth and yields [129].

In addition to helping farmers manage nutrient composition, crop rotation improves physical soil structure and properties. Differing root structures impact soil porosity, aeration, and water-holding capacity [130]. Fibrous-rooted crops such as grasses hold soil particles together, preventing erosion, whereas crops with tap root systems, such as carrots and radishes, break compacted soil, boosting drainage and root penetration [8]. Carrots (*Daucus carota* L.) can be used effectively in crop rotation with crops such as barley, as their taproots can improve soil structure and provide nutritional and economic benefits [131–133].

4.2. Nutrient Optimisation

Nutrient optimisation is another advantage of crop rotation systems that requires prioritisation. Legumes effectively fix nitrogen from the atmosphere for use by other crops. While all plants require nitrogen, cereal crops such as wheat or corn have a higher demand for nitrogen compared to legumes. In traditional rotations, pea or clover, which are legumes, are followed by cereal crops. This integration ensures that the nutrient balance is intact, decreasing the use of synthetic nitrogen fertilisers and preventing soil degradation [21,134].

Crop rotation also enhances the nutrient-cycling ability of soil microbial communities [135]. The key factor that supports this process is the soil health score, which provides a comprehensive assessment of soil quality and its capacity to support crop production without degradation [136,137]. Soil health scores are usually calculated using a combination of physical, chemical, and biological indicators, such as soil texture, organic matter content, nutrient level, and microbial composition [138,139]. Cereal crops such as wheat and corn, for example, are heavy users of nitrogen and can exhaust soil nitrogen reserves if grown continuously in monocropping systems, thus requiring applications of nitrogen-rich fertilisers in between seasons. Rotating these crops with legume crops enhances soil fertility by restoring nitrogen through biological nitrogen fixation. Leguminous plants are symbionts with nitrogen-fixing bacteria (rhizobia) in their root nodules, which fix atmospheric nitrogen into forms readily available for future crops, obviating the use of synthetic fertilisers [14,140]. Thus, crop rotation decreases the use of synthetic nitrogen fertilisers and enhances soil fertility sustainably.

As shown in Figure 1, nitrogen savings are dependant on crop type, rotation structure, baseline soil fertility, and the inclusion of nitrogen-fixing legumes.

Crop rotation enhances carbon sequestration as different crops make varying contributions to SOM [146]. Stored carbon is essential for sustainable farming as it improves soil aggregation, stabilises soil structure, and decreases carbon emissions owing to the reduced amount of applied fertiliser [147]. In addition to nitrogen, crop rotation optimises the availability of other major nutrients such as phosphorus and potassium. Deep-rooted crops such as sunflowers and alfalfa bring nutrients from deeper soil to the surface through absorption

by the roots, depositing them near the surface when the plant undergoes decomposition. These nutrients are then made available to shallow-rooted crops in subsequent cycles. This prevents nutrient stratification and improves soil nutrient cycling. Deep root systems also contribute to enhancing organic matter accumulation by depositing root litter, which is beneficial to soil microbial activity, enhancing decomposition and nutrient uptake [148,149].

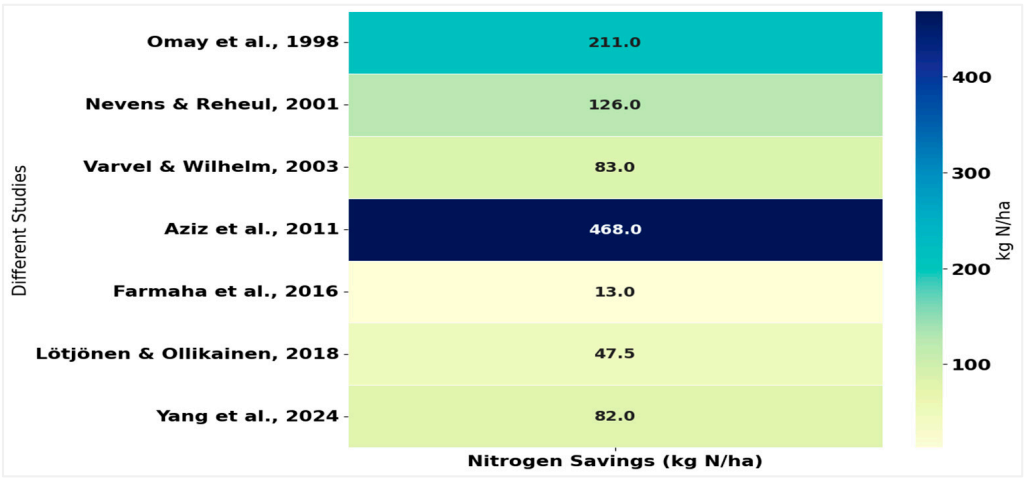


Figure 1. Heatmap of reported nitrogen savings from crop rotations compared to monoculture, based on the field data gathered from different studies [109,115,141–145]. This graph represents a summary of field-based studies quantifying nitrogen savings achieved through crop rotation. The heatmap presents a combination of data from eight sources, highlighting the reduction in nitrogen fertiliser inputs or increases in total soil nitrogen rates across different crop systems.

Crop rotation also minimises the risk of nutrient imbalances that are a result of monoculture practices. Monoculture tends to deplete nutrients used intensively by the main crop, such as nitrogen when growing grains, while other nutrients, such as phosphorus, accumulate to levels that can be detrimental to plant growth in some forms [150,151]. Through crop rotation, farmers can achieve a more balanced use and distribution of soil nutrients. For example, by rotating crops that require high levels of phosphorus and potassium for growth, such as root crops like carrots and turnips, farmers can improve bioavailability of phosphorus and potassium [26]. In addition, the use of cover crops such as clover, vetch, or rye in the rotation cycle enhances nutrient retention by taking up nutrients that might be lost in the soil and reducing leaching losses through the root structure, while retaining nutrients within the roots. Cover crops accept excess nutrients that would otherwise be lost in runoff and make them available for subsequent crops through decomposition, which releases those nutrients back into the soil. Cover crops also contribute to SOM, which enhances the nutrient-holding capacity and structure of soil, thereby decreasing the need for external fertilisers [152–154].

As shown in Figure 2, crop yields in rotation systems comprising soybean and corn were higher than in continuous cropping, underscoring the advantages of rotational techniques for productivity [155].

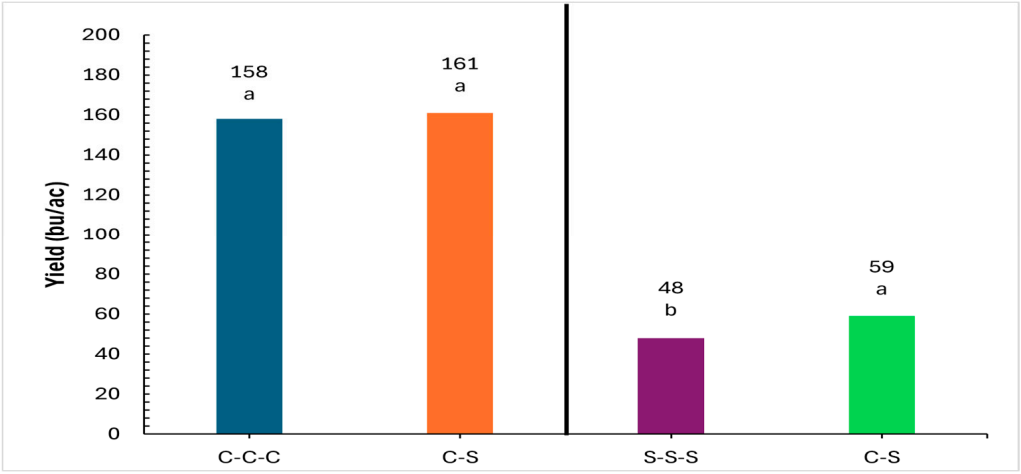


Figure 2. Case study example of crop yields using monoculture and crop rotation practices for 2000–2001 (measured in bushels per acre, bu/ac). C-C-C, continuous corn; C-S, corn–soybean, S-S-S, continuous soybean. Each value above a bar represents the mean. Different letters denote significant differences between rotation systems ($p < 0.05$) [155].

4.3. Pest and Disease Control

Crop rotation is a key component of integrated pest management (IPM) as it reduces reliance on chemical pesticides [21,156]. By altering crops, this system disrupts the life cycles of pests and pathogens [157] and enhances agrobiodiversity, thereby enhancing natural pest control through beneficial organisms such as predatory insects and soil microbes [158–160].

Disrupting the availability of host plants for pests and conditions for pathogens that cause disease effectively reduces pest incidence [26]. For instance, reducing pest abundance can be achieved by intercropping plants that are prone to diseases or attractive to different insects with less vulnerable plants. Rootworm larvae that feed on corn can be reduced by interplanting corn with beans. These works well since the larvae usually spend much of their life cycle on corn roots, which leaves them exposed when corn is switched out for a non-host crop like beans [21,161].

Additionally, crop rotation can benefit soil health by reducing the incidence of soil-borne diseases, as crop diversity may decrease soil pathogens through promoting beneficial microorganisms and improving crop disease management [162]. For instance, Kelley et al. [163] demonstrated that crop rotation can reduce the incidence of disease through suppressing the populations of soybean cyst nematode (SCN) and increasing soybean yield. In the twenty-year field study in Kansas, they showed that continuous soybean monoculture resulted in the highest SCN populations and the lowest yields, while crop rotations incorporating wheat or grain sorghum not only decreased SCN populations but also increased seed weight and chemical properties of the soil such as nitrogen and total carbon content. The study illustrates how crop rotation minimises disease incidence and supports both long-term soil productivity and plant health. This clearly demonstrates that crop rotation effectively reduces disease pressure caused by the soybean cyst nematode. When soil quality is improved through crop rotation practices, it can support various beneficial organisms that control or suppress soil-borne pathogens [164].

In addition to these direct advantages, the cropping system reduces reliance on chemical pesticides, contributing to more sustainable agriculture [165,166]. As mentioned previously, low pesticide application reduces the levels that can leach into the environment, thus protecting all wildlife, including beneficial fauna that act as natural insect repellents and terrestrial and aquatic species that can be negatively impacted by pesticides [167,168]. Therefore, crop rotation can be a practical component of a broader integrated pest man-

agement approach. Crop rotation ensures healthy crop growth and minimises problems associated with destructive pests in agricultural practices.

4.4. Weed Control

Organic weed control is another benefit of crop rotation, as complex crop systems do not require herbicides to manage weeds [169,170]. Monoculture systems provide ideal environments for certain weed species to thrive. However, crop rotation disrupts this, as the crops have varying growth characteristics, including differences in height, rate of development, competitive ability, and root and canopy architecture, depriving weeds of a stable environment [171–173].

For instance, planting cover crops with high biomass production or with a high tillage rate can effectively suppress weeds and prevent light-dependent weed seeds, such as lambsfoot and red pigweed, from germinating or establishing. Clover and alfalfa are legumes that can fix nitrogen, improve soil health, and promote vigorous crop growth, which increases competition with weeds for resources such as light and nutrients. Furthermore, some crop species secrete compounds that affect weed seed germination, such as the allelochemicals released by rye and sorghum [51].

Table 3 presents the broader benefits and effectiveness of crop rotations and related agronomic practices on weed management. The table is based on field studies demonstrating how rotation, intercropping, and cover cropping contribute to weed control.

Rotational crops such as rice–wheat and cotton–wheat reduce weeds; however, some nutrients in the soil may be depleted over time. Like other legumes, mung bean adds nitrogen to the soil, increasing subsequent wheat yields [174]. Sorghum is efficient in weed suppression through *allelopathy*, a phenomenon in which the plant releases biochemicals that effect the growth, survival, and reproduction of other organisms; however, if sorghum residues are not managed properly, they can inhibit wheat germination [175,176]. The choice of rotation depends mainly on the regional conditions and specific objectives, such as increasing yield, controlling weeds, or improving soil quality [177].

Table 3. Effects of different crop rotation systems and related agronomic practices in weed suppression, based on field study data.

Crop Rotation or Related Practices	Effect on Weed Management	Reference
Maize–soybean rotation	Weed seed density reduced by ~80% (450 seeds/m^2) vs. monoculture (2250 seeds/m^2).	[100]
Wheat, maize, cereals with legumes and vegetables Monoculture vs. crop rotation and intercropping.	Reduced weed density in 75–78% of comparisons.	[171]
Field pea–barley monoculture vs. intercropping	Suppressed 96% of broadleaf weeds.	[178]
Multiple systems (rice–wheat, cotton–wheat, sorghum–wheat, mung bean–wheat, intensive and multiyear rotations)	Effective in reducing specific weeds; some pest challenges remain. Suppresses weeds but allows pest carryover. Reduces weed infestation effectively; suppression is attributed to allelopathy. Moderate weed control reduces herbicide reliance. Significant weed suppression: wheat (−51%), maize (−70%), cotton (−66%). Reduces weeds in cotton by 31–57% through extended cropping cycles.	[179]
Rice–wheat with residue management (mulching/incorporation vs. no residue)	Weed biomass reduced by up to 31.3%.	[174]
Rye, maize, soybean, cotton monoculture vs. cover-crop-based rotation	Reduced broadleaf weeds by 96% and grass weeds by 61%.	[51]

4.5. Increased Crop Yields

Crop rotation contributes to food security and better soil health by raising crop yields and improving agricultural sustainability. Crop rotation is more productive in the long run because it improves soil fertility and can result in safer agricultural production by lowering dependency on synthetic pesticides compared to monoculture. It reduces the need for intensive soil management programmes, disrupts pest and disease life cycles, and improves general soil health [26,180,181].

Crop rotation can enhance yields by improving nutrient availability in the soil, particularly through nitrogen fixation. The yield impact varies depending on the crop species, soil, and climate [182,183]. Legumes benefit subsequent cereal crops such as maize and wheat, although outcomes depend on factors such as nutrient and water management [184,185]. Different crops have different nutrient requirements and effects on soil fertility.

In diversified systems, yield gains also partly resulted from optimised root–shoot dynamics. In cotton-based intercropping, enhanced root system development further supports higher photosynthesis activity and nutrient uptake, thereby contributing to improved biomass and yield stability [84]. Root–shoot signalling further plays a crucial role in enhancing stress resilience and resource-use efficiency [91,92].

A comparison of crop-by-crop yields indicates that crop rotation systems have higher yields than those of monoculture systems. For example, rotating maize with soybeans showed a 13–15% yield increase compared to continuous maize monoculture [186]. Cropping wheat after legumes increased yields by 13–31.71%, depending on the intensity of the rotation and the legume harvest [187]. Similarly, two-year maize–wheat–soybean rotations boosted soybean yields by 39–44% [188], while annual rotations increased maize and soybean yields by 20% and 22%, respectively [189].

Maize has shown positive yield responses to rotation. Meta-analyses and long-term trials have consistently reported an increase in maize yields in rotated systems compared to continuous monoculture, with crop rotation largely increasing agricultural production [182]. Greater benefits have been observed when legumes are included in the sequence, with meta-analyses showing yield advantages in succeeding maize of up to 34% when using legume crops compared to non-legume predecessors [134,190]. A global systematic review with a meta-analysis revealed significant yield advantages of legume-based rotations for maize [191]. These benefits are linked to improved nitrogen availability from legume nitrogen contribution [100] and limited pest incidence, as continuous growing of maize over within several years increases pest pressure compared to well-planned rotation systems. Crop rotation provides long-term yield and stability benefits for maize production, according to long-term research [192].

Figure 3 presents findings of 19 field studies reporting the percentage yield outcomes of crop rotation compared to monoculture. The data indicate that yield improvements associated with crop rotation range from modest (3%) to substantial (up to 100%), depending on factors such as crop type, rotation varied, fertiliser rate, agricultural practices, and geographical region.

These findings demonstrate that, in contrast with continuous cropping under both no-till and chisel tillage, soybean yields were higher in rotated systems. Compared to moldboard ploughing, wheat fared better under chisel and ridge tillage. Alfalfa and wheat straw yields did not significantly differ between treatments [155].

Table 4 lists the effects of tillage and rotational cropping systems on crop yields based on the findings of [155]. The data indicates the differences in monoculture (monocrop or continuous cropping) and rotational cropping systems across tillage treatments. The findings show that crop rotation can promote stability of the yield and reduce the need to use intensive tillage operations.

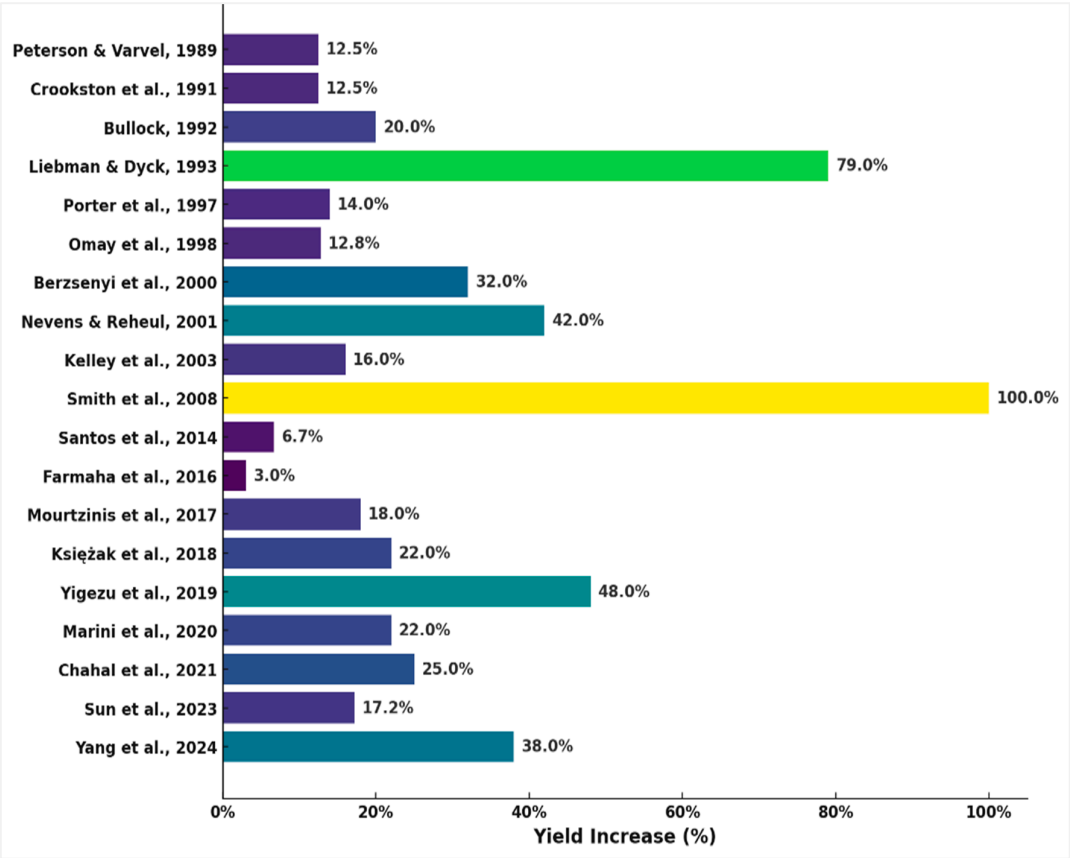


Figure 3. Reported yield increases from crop rotation compared to monoculture: a chronological review of field studies (1989–2024) based on data from [100,109,117,119,135,141,142,145,163,171,183,186,189,193–198]. Colour categories indicate the percentage range of yield increases: dark purple (0–10%), purple (11–20%), blue (21–30%), teal (31–50%), green (51–80%), and yellow (81–100%).

Table 4. Case study example of comparative yield analysis (crop rotation vs. monoculture), based on field data gathering from Singer et al. [155].

Crop Type	Treatment (Continuous/ Monoculture)	Yield (kg/ha, Monoculture)	Treatment (Rotated)	Yield (kg/ha, Rotated)	% Difference
Soybeans (2001)	No-till treatment (continuous)	3701	W/S-C-S rotation	3564	−3.6%
Soybeans (2001)	Chisel treatment (continuous)	2960	W/S-C-S rotation	3228	9.10%
Wheat (2001)	Mouldboard plough	Higher by 8%	Chisel/ridge tillage	−8%	-
Alfalfa (2000/2001)	Mouldboard plough	No significant difference	No-till	0%	-
Wheat straw (2000/2001)	Chisel tillage	~1000 kg/ha	No-till	~1000 kg/ha	-

5. Environmental Benefits

5.1. GHG Emissions

Agricultural CO_2 emissions mainly result from soil disturbance (e.g., tillage), crop residue decomposition, and fossil fuel use in field operations. Ploughing accelerates microbial respiration and CO_2 release, while synthetic nitrogen fertilisers add indirect emissions via energy-intensive production [8,15,199].

Crop rotation mitigates CO_2 emissions by enhancing SOM and reducing the need for tillage and fertilisers. Systems including legumes and perennials improve carbon inputs and sequestration over time [16,200].

Other GHGs are also affected. Legume-based rotations can reduce N_2O emissions by improving nitrogen efficiency, with reductions up to 39% [23]. In flooded rice systems, rotating with upland crops such as maize or sorghum lowers CH_4 emissions by up to 84% by disrupting anaerobic soil conditions [201].

Agriculture contributes significantly to GHG emissions, including carbon dioxide, methane, and nitrous oxide, mainly due to intensive farming, artificial fertilisers, mechanisation, crop residue decomposition, and specific practices such as rice cultivation and nitrogen fertilisation [202,203]. Hashim et al. [204] highlighted how agricultural practices, including pest control, tillage, and nitrogen application, affect carbon dioxide emissions. By having a clearer understanding of these sources of GHG emissions, effective climate change mitigation strategies can be implemented [205]. The following section explores how crop rotation can reduce GHG emissions, particularly soil carbon dioxide emissions.

Beyond reducing GHG, crop rotation improves soil health, biodiversity, and nutrient use while lowering erosion, leaching, and agrochemical dependence.

5.2. Carbon Dioxide Emissions

Crop rotation improves water use efficiency by enhancing soil structure, increasing infiltration, and reducing runoff and evaporation. Systems with legumes or deep-rooted crops help retain moisture and reduce irrigation needs, supporting drought resilience [14,206].

Crop rotation boosts soil carbon storage, reducing CO_2 emissions [207]. Mixed grass–legume pastures enhance soil organic carbon (SOC) content, while monoculture depletes it [208]. The use of legumes in cropping systems, particularly as cover crops, can reduce the need for synthetic fertiliser use, lowering related emissions. Moreover, legume cover crops are a reliable means to enhance cash crop yields. Legume rotations, such as clover–wheat, also reduce nitrogen runoff and GHG emissions [115,209,210]. The benefits of crop rotation compared to monoculture on environmental with regarding the need to fertilisation, nitrogen runoff, and GHG emissions, are shown in Figure 4.

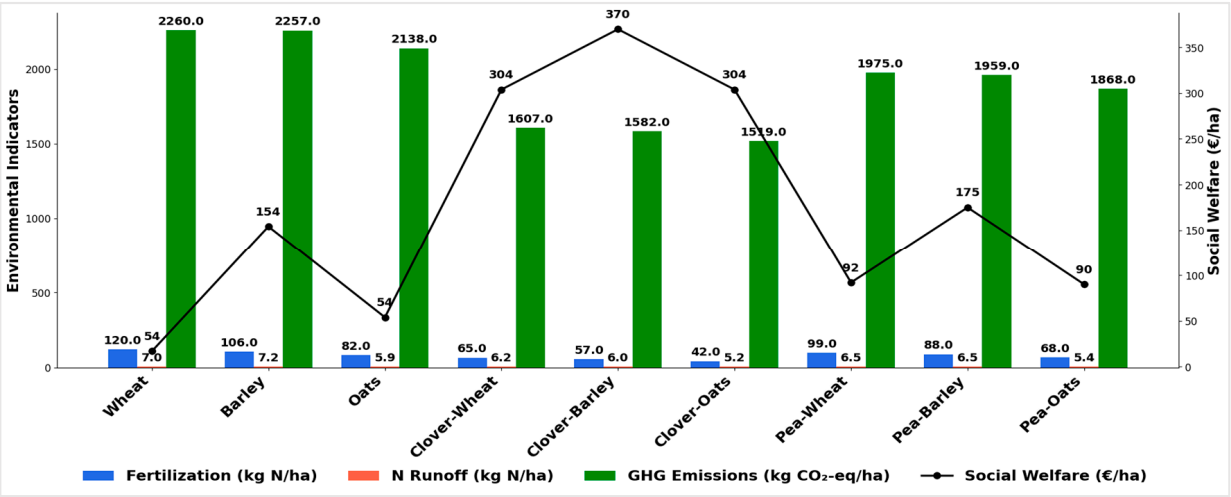


Figure 4. Case study example of environmental and economic gains of crop rotation over monoculture: cost–benefit comparison of GHG emissions, fertilisation, nitrogen runoff, and social welfare. The figure contrasts monocultures and legume-based rotations (clover, peas) in terms of environmental impacts and economic returns. The data derived adopted from [115].

General principles emphasises that crop rotation benefits agriculture. However, numerous studies have examined the benefits of certain crop rotation systems and management techniques in terms of GHG emissions. Some of these important findings can be summed up as follows.

Sainju et al. [15] examined the impact of various crop rotation systems and nitrogen fertilisation rates on soil CO_2 emissions and carbon balance over a 4-year period. Their study aimed to evaluate the effects of crop rotation and nitrogen levels on soil CO_2 and CH_4 emissions. They achieved this by investigating spring wheat monocultures and spring wheat–pea rotation under different nitrogen levels (0, 50, 100, and 150 kg/ha, respectively). Soil CO_2 emissions were observed to peak following planting, fertilisation, and heavy rains. It was determined that these emissions were influenced by the treatments, i.e., nitrogen fertilisation and crop rotation, which were designed to evaluate their impacts on soil CO_2 and CH_4 emissions. Crop rotation can influence cumulative yearly CO_2 emissions, depending on the year and nitrogen application rate; this variability is due to factors such as the effect of the fertiliser rate of N having a differential impact on soil CO_2 emissions under varying conditions and fluctuating weather patterns influencing carbon inputs and outputs. Yields relative to gas flux, which point to the rate of emission of CO_2 among the soil and atmosphere, did not differ between treatments but varied annually [15,211]. The carbon balance was consistently negative, regardless of treatment. A legume and non-legume rotation with less nitrogen reduced soil CO_2. Spring wheat systems release carbon in semiarid regions [15].

Rigon et al. [16] compared the short-term soil CO_2 emissions from two no-till experiments with various winter and spring crops, specifically pearl millet, Sunn hemp, and forage sorghum as spring crops and ruzigrass, sunflower, triticale, and grain sorghum as winter crops, followed by summer soybean rotations. The average CO_2 emission rate and total CO_2 emissions over the growing season were reported for both soil types. One experiment found that planting Sunn hemp as a spring crop resulted in 12% lower CO_2 emissions and a 9% higher soybean yield than pearl millet. The other experiment showed a 17% reduction in CO_2 emissions when ruzigrass and sorghum were grown in succession in winter while maintaining soybean productivity at levels comparable to other treatments. Soil moisture, crop residue nitrogen, and clay content were identified as factors influencing CO_2 emissions levels.

Yang et al. [212] evaluated and compared the carbon footprints of five cropping systems in the North China Plain from 2003 to 2010. The crop rotations included different combinations and lengths; these combinations consisted of various sequences of crops, including sweet potato, cotton, winter wheat, summer maize, ryegrass, and peanut, rotated over different year spans. The carbon footprint was analysed based on area, biomass, and economic output. The system with the lowest annual CO_2 emission footprint was sweet potato–cotton–sweet potato–winter wheat/summer maize, representing 27.9–28.2% of the highest-emitting winter wheat/summer maize rotation. In order of decreasing carbon footprint, the five cropping systems, ranked from the lowest to highest, were (1) rotation of sweet potato and cotton in alternation, followed by winter wheat with summer maize; (2) cotton monoculture; (3) ryegrass and cotton with peanut, then winter wheat with summer maize; (4) peanut with winter wheat and summer maize; and (5) continuous winter wheat with summer maize.

Maas et al. [17] continuously monitored CO_2 fluxes over 2 years in fields with newly established perennial forage crops (mainly alfalfa and timothy grass) with annual spring wheat and rapeseed crops planted in adjacent areas. The perennial forage absorbed approximately double the atmospheric CO_2 that yearly crops did, at 4480 $\pm$ 1840 kg C/ha compared with 2470 $\pm$ 700 kg C/ha, respectively. By factoring in harvested materials

and calculating total GHG emissions—including both direct CO_2 emissions and other GHGs like N_2O—converted into CO_2 equivalents (CO_2 eq) based on their global warming potential, perennial forage had a net sink of 8470 ± 5640 kg CO_2 eq/ha, whereas annual crops released 3760 ± 2450 kg CO_2 eq/ha. Continuous CO_2 measurements showed that perennial systems sequestered significantly more atmospheric CO_2 and acted as more robust CO_2 sinks than annual rotations [17].

Abagandura et al. [18] measured CO_2 fluxes under a spring wheat–cover crops–maize–pea/barley–sunflower crop rotation system and continuously grown spring wheat over 2 years in North Dakota. The crop rotation system progressed through the following sequence: spring wheat, followed by cover crops, then maize, followed by either pea or barley, and concluded with sunflower. They found that CO_2 fluxes were generally lower in 2016 than in 2017 and attributed this difference to high soil temperatures in 2017, which increased microbial activity and carbon mineralisation. Within each year of the study, increasing soil temperature positively correlated with higher measured CO_2 flux. Although daily CO_2 flux levels varied between crop phases, cumulative annual fluxes did not differ significantly. This indicates that the rotational diversity of crops has a limited influence on soil temperature and microbial processes that drive GHG emissions. However, soil temperature appeared to be the primary factor influencing the variations in CO_2 flux levels between years and crop growth phases. According to this study, rotational diversity, residues, or SOC content did not significantly affect CO_2 emissions.

In South Dakota, Wegner et al. [213] reported several practices that could impact CO_2 emissions from soils in a corn–soybean cropping system, such as leaving crop residues or residual retention in the field after harvest, which resulted in lower cumulative CO_2 emissions than residue removal. Additionally, the use of cover crops decreased cumulative CO_2 emissions in over two successive years relative to treatments without cover crops. When the cover crops were planted in both corn and soybean rotation systems, CO_2 fluxes were significantly lower than in treatments without cover crops. Retaining residues and incorporating cover crops into the system provided notable benefits for mitigating soil CO_2 emissions.

Behnke et al. [214] revealed that maize is associated with soil conditions and fluxes more conducive to higher CO_2 emissions than soybean. This study was implemented as a long-term cropping system experiment in Illinois, USA. Moreover, it investigated how yields, soil characteristics, and GHG emissions relate to different cropping systems, such as corn monocultures and corn–soybean rotations. The researchers discovered that while corn in the rotation released more N_2O than soybeans, corn was associated with soil conditions and fluxes more conducive to higher CO_2 emissions than soybeans. This finding is consistent with other research highlighting the impact of crop rotations on GHG emissions and soil properties [214].

Alluvione et al. [215] demonstrated that plant species within a rotation in Northeastern Colorado affected CO_2 emissions. Barley (maize–barley rotation) emits a higher annual cumulative CO_2 flux than maize (continuous maize) and dry bean (maize–dry bean rotation). These variations in the cumulative CO_2 flux were attributed to the quantity and quality of the decaying residues from the preceding crops.

In a long-term field study from 2007 to 2019, Wang et al. [216] compared three tillage methods—no-till, subsoiling tillage, and mouldboard ploughing—in a farming system in which crop residues were returned to the soil. They measured CO_2 fluxes from soils receiving different tillage treatments. Their principal findings regarding CO_2 emissions showed that no-till and subsoiling tillage had lower fluxes than mouldboard ploughing over a 2-year average. Specifically, no-till reduced the average CO_2 flux by 14.5% when

compared with mouldboard ploughing, whereas subsoiling tillage reduced the average flux by 8.5% when compared with mouldboard ploughing.

Nyambo et al. [217] examined the effects of tillage, crop rotation, and residue handling on the soil's CO_2 flux. Specifically, they assessed conventional tillage versus no-till, maize, oats, vetch rotations, residue retention, removal, and biochar addition. These factors were evaluated individually and in combination with each other. For example, zero-tillage with residue retention, conventional cultivation with residue removal, and zero-tillage with biochar addition were considered. The CO_2 flux was measured under different treatment combinations. They showed that tilling increased fluxes over no-till by approximately 26.3%. Residue removal yielded lower fluxes than either retention or biochar addition. Across most of the agricultural methods tested, the CO_2 fluxes were generally higher in summer than in winter [217–219]. No-till maintains more carbon in the topsoil than conventional tillage. Biochar leads to a lower total carbon content than residue practices. Thus, no-till and residue removal reduced soil CO_2 emissions under most agricultural methods evaluated [217].

Conservation tillage methods, such as minimum and no-tillage, can help lower GHG emissions compared to traditional tillage [8]. The System Approach to Land Use Sustainability (SALUS) simulation model showed that compared with conventional tillage, minimum tillage reduced SOC content losses by 17%, and no-tillage reduced losses by 63% over 15 years. Implementing conservation tillage techniques decreases carbon emissions from farming. A synergistic relationship exists between conservation tillage (especially no-tillage) and precision agriculture strategies that can reduce total CO_2 emissions by 56% compared with the CO_2 emission reduction in conventional tillage. It should be noted that this review does not account for emissions from agricultural machinery, such as fuel use for tillage and harvesting, which can significantly contribute to the total carbon footprint of farming systems [23,220].

5.3. Methane Emissions

Methane emissions (CH_4) are particularly significant in rice-based cropping systems owing to anaerobic conditions that promote methanogenesis, as methanogenic archaea, which generate methane, thrive in oxygen-deprived conditions. The soil in flooded rice paddies becomes oxygen-depleted, which creates an ideal environment for these microorganisms to flourish [221]. In addition, alternate wetting and drying irrigation management lowers CH_4 emissions from paddy fields [221–223]. The effectiveness of crop rotation in reducing CH_4 emissions is demonstrated through research comparing different cropping systems. As shown in Figure 5, replacing continuous rice cultivation (RR) with corn–rice or sorghum–rice rotations reduce annual methane emissions by approximately 78–84%, depending on crop type and seasonality, while also impacting nitrous oxide and carbon dioxide fluxes.

5.4. Nitrous Oxide Emissions

Leguminous crop rotations enhance nitrogen fixation, decreasing the reliance on synthetic fixed nitrogen sources, which contribute to one of the primary sources of N_2O emissions [224]. However, improper management of leguminous residues can increase N_2O emissions, making nitrogen management practices essential [225]. Diversified crop rotations improve soil nitrogen cycling efficiency and reduce nitrogen leaching, thereby reducing overall N_2O emissions [141]. The impact of different crop rotation on N_2O emissions is shown in Figure 6, which compares the emissions from different rice-based rotations over a 2-year period.

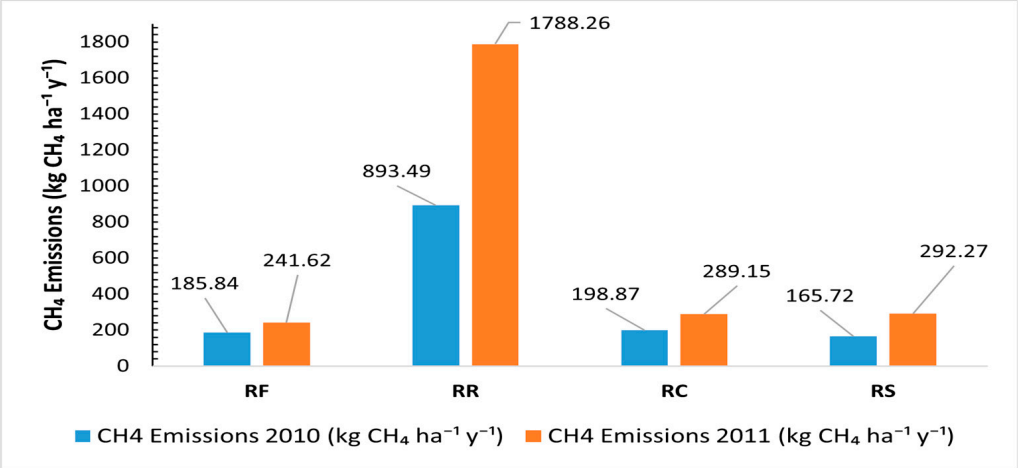


Figure 5. Case study example of annual methane (CH_4) emissions (kg CH_4 ha^{-1} y^{-1}) in different crop rotation trends for 2010 and 2011. RF, rice–fallow; RR, continuous rice; RC, rice–corn; RS, rice–sorghum. Based on field data adopted from [201].

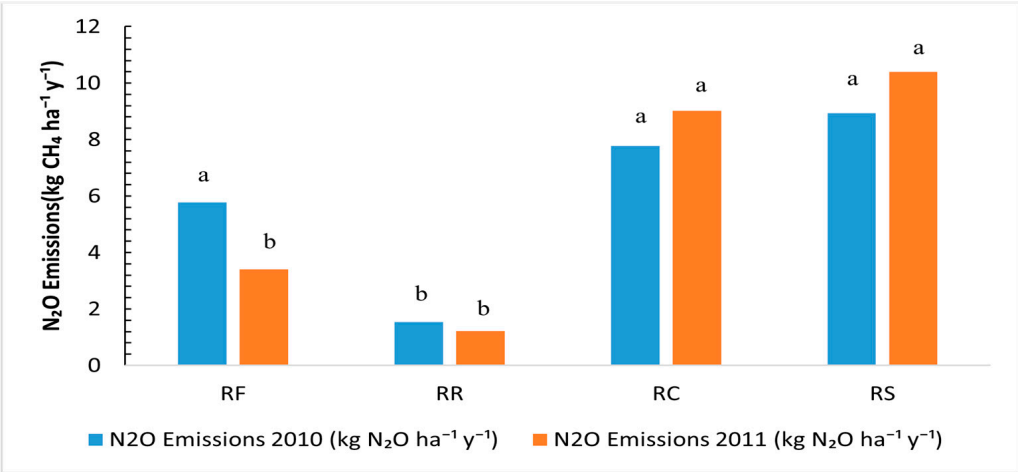


Figure 6. Case study example of nitrous oxide (N_2O) emissions in different rice-based crop rotations over a 2-year period. The data here concur with the main findings of Cha-un et al. [201] by emphasising the relative emission from each type of crop rotation. Different letters denote statistically significant differences ($p < 0.05$) between treatments. RF, rice–fallow; RR, continuous rice; RC, rice–corn; RS, rice–sorghum. Based on field data adopted from Cha-un et al. [201].

The above discussion highlights the potential of crop rotation to decrease GHG emissions, particularly CO_2, through carbon sequestration and reduced fertiliser use. However, its impact on methane and nitrous oxide differs depending on the rotation type, management, and environment. Holistic design is essential to reduce overall GHG emissions. For example, alternating wetting and drying in rice–rice rotations decrease methane emissions, while proper nitrogen management in legume–sorghum rotations helps limit nitrous oxide emissions.

6. Conclusions

Crop rotation is essential in farming systems for improving soil health, pest and weed control, and nutrient cycling. Optimising crop sequences, including legumes, cereals, and root crops, can help regulate soil fertility, improve nutrient availability, and control pest and disease patterns. Furthermore, cover crops and rotational grazing can be practiced in conjunction with crop rotation to promote practices consistent with organic farming,

thereby improving soil structure, limiting chemical inputs, and increasing biodiversity in agricultural production systems.

Beyond its agronomic advantages, crop rotation plays a crucial role in environmental sustainability. It contributes to soil carbon sequestration, reduces GHG emissions, and minimises chemical runoff, thereby supporting global efforts to combat climate change. Documented outcomes include increased crop yields after legume implementation, improved soil health that led to increased SOC content, and decreased nitrogen input requirements, as well as reduction of N_2O emissions up to 39%. Crop rotation enhances nitrogen fixation and nutrient cycling, thus reducing the need for synthetic fertilisers. This enhancement contributes to environmental protection and reduced production costs. Moreover, crop rotations increase yields, promotes long-term soil health, improves climate resilience and water retention, and minimises reliance on chemical inputs. Therefore, crop rotation acts as a multidimensional solution ensuring enduring soil fertility and food security.

Nevertheless, the study has some limitations. First, most of the research used is focused on temperate regions such as Europe and North America, which limits the possibility of generalisability in tropical, arid, and highland regions where climate, soil, and crop diversity differ significantly. Second, information from case studies and empirical findings on crop rotation practice adoption and outcomes among smallholder and subsistence farmers are limited, particularly in low- and middle-income countries. These farmers often face unique constraints such as limited access to inputs, and extension service could respond differently to the application of rotation systems. Therefore, future research should include more studies across different geographies and focus on small farmers to ensure that crop rotation strategies are more useful for everyone.

Despite its advantages, promoting crop rotation through policy remains challenging. Monoculture practices continue to be used in many agricultural regimes, with a focus on their short-term profitability. Their major constraints include the lack of subsidies, limited extension support, and weak market support for rotational crops. Farmers cannot easily employ rotation systems without any financial or technical assistance. Addressing such gaps requires coordinated policies, institutional support, and the convergence of climate and food security goals.

Future research directions should focus on the following key areas:

- Region-specific rotation schemes: Developing and testing rotations tailored to arid, tropical, and mountainous environments, where existing data are limited;
- Integration into precision agriculture technologies: Examining how remote sensing and machine learning can be used to optimise rotation schedules, nutrient management, and yield productivity;
- Long-term ecological impact quantification: Conducting multiyear experiments to assess biodiversity trends, carbon storage, and GHG fluxes under diverse rotation regimes;
- Economic and policy modelling: Evaluating the cost-effectiveness of rotation strategies under various policy settings (e.g., carbon pricing, subsides, conservation incentives);
- Integration of new crop species: Exploring the integration of non-conventional or climate-resilient crops (i.e., pigeon pea, quinoa, or native legumes) into rotation systems to enhance system diversity and ecological functionality;
- Farmer-led innovation and socio-cultural adaptation: Considering social acceptability, incorporation of indigenous knowledge, and participatory practices when developing crop rotation systems, especially among smallholder farmers and subsistence farmers.

By acknowledging the above limitations, further research is required to enhance crop rotation practices, implement emerging technologies, and develop region-specific strategies that optimise agricultural productivity as well as environment resilience. The evidence

resulting from previous studies confirms that crop rotation is a proven method to promote resilience, ecological health, and economic viability in modern agriculture.

7. Policy Recommendations

Promoting crop rotation as a climate mitigation tool requires coordinated policy action. Governments should support diversified cropping through subsidies, carbon credits, or tax incentives tied to soil health and emissions goals. Given that farmers cannot easily employ rotation systems without financial or technical assistance, strengthening extension services and providing region-specific training—especially for smallholders—can improve adoption. Integrating crop rotation into national climate plans (NDCs), certification schemes, and public procurement can enhance uptake, while public–private partnerships can scale locally adapted solutions. These interventions are justified by crop rotation's ability to reduce N_2O emissions by 39% [109] and boost yields by 13–44% [163].

Author Contributions: Conceptualization, Z.K.A.-M., I.M.K. and V.V.; data curation, Z.K.A.-M., I.M.K. and V.V.; formal analysis, Z.K.A.-M. and I.M.K.; investigation, Z.K.A.-M., I.M.K. and V.V.; methodology, Z.K.A.-M. and I.M.K.; resources, Z.K.A.-M. and I.M.K.; software, Z.K.A.-M. and I.M.K., supervision, I.M.K. and V.V.; validation, Z.K.A.-M., I.M.K. and V.V.; visualization, Z.K.A.-M., I.M.K. and V.V.; writing—original draft, Z.K.A.-M.; writing—review and editing, Z.K.A.-M. and I.M.K. All authors have read and agreed to the published version of the manuscript.

Funding: The research was funded and supported by the TechCoach (Project ID: 101182908; NRDI ID: 2020-2.1.1-ED-2024-00342) and CSR (Project ID: 101216573; NRDI ID: 2025-3.1.2-KÖA-2025-00020) Horizon Europe projects.

Conflicts of Interest: The authors declare no conflicts of interest.

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